

Exercises: Gradient algorithms

Exercise 1 (A quadratic example in $\mathbb{R}^2$). Consider, for $\gamma > 0$, $f(x) = \frac{1}{2}(x_1^2 + \gamma x_2^2)$. We apply the gradient descent method with optimal step, starting at $x^{(0)} = (\gamma, 1)$.

1. Show that f is m -convex with M -Lipschitz gradient. Find the tightest m and M constants.
2. Show that

$$x^{(k)} = \left(\gamma \left(\frac{\gamma-1}{\gamma+1} \right)^k, \left(-\frac{\gamma-1}{\gamma+1} \right)^k \right)$$

and

$$f(x^{(k)}) = \frac{\gamma(\gamma+1)}{2} \left(\frac{\gamma-1}{\gamma+1} \right)^{2k} f(x^{(0)})$$

3. Show that, on this example, the convergence is exactly linear, that is $f(x^{(k)}) - v^\sharp$ is a geometric series. Give its reason. Compare with the theoretical bound.
4. When is this algorithm fast and slow ?

Exercise 2 (Strongly convex - optimal step). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a m -convex $\mathcal{C}^2$ function. Define, for given $x^{(0)}$,

$$\begin{aligned} \tilde{f}_k : t &\mapsto f(x^{(k)} - t \nabla f(x^{(k)})) \\ t^{(k)} &= \arg \min_{t \in \mathbb{R}} \tilde{f}_k(t) \\ x^{(k+1)} &= x^{(k)} - t^{(k)} \nabla f(x^{(k)}) \end{aligned}$$

1. Show that there exists $M \geq m$ such that $mI \preceq \nabla^2 f(x^{(k)}) \preceq MI$
2. Show that, for any interesting t (to be defined) we have

$$\tilde{f}_k(t) \leq f(x^{(k)}) - t \|\nabla f(x^{(k)})\|_2^2 + \frac{Mt^2}{2} \|\nabla f(x^{(k)})\|_2^2$$

3. Show that,

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2M} \|\nabla f(x^{(k)})\|_2^2$$

4. Show that $f(x^\sharp) \geq f(x) - \frac{1}{2m} \|\nabla f(x)\|_2^2$
5. Show that

$$f(x^{(k+1)}) - f(x^\sharp) \leq \left(1 - \frac{m}{M}\right) [f(x^{(k)}) - f(x^\sharp)]$$

6. Show that the algorithm converges, and give its convergence speed.

Exercise 3 (Strictly convex case). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a $\mathcal{C}^2$, convex function with M -Lipschitz gradient such that $f(x^\sharp) = \inf f$. We define, for given $x^{(0)}$,

$$x^{(k+1)} = x^{(k)} - t \nabla f(x^{(k)})$$

with $t \leq 1/M$.

1. Show that, for all x and y $(y-x)^\top \nabla^2 f(z)(y-x) \leq M \|y-x\|_2^2$
2. Show that

$$f(y) \leq f(x) + \nabla f(x)^\top (y-x) + \frac{M}{2} \|y-x\|_2^2$$

3. Show that $f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{t}{2} \|\nabla f(x^{(k)})\|_2^2$
4. Show that

$$f(x^{(k+1)}) \leq v^\sharp + \nabla f(x^{(k)})^\top (x^{(k)} - x^\sharp) - \frac{t}{2} \|\nabla f(x^{(k)})\|_2^2$$

5. Deduce that

$$f(x^{(k+1)}) \leq v^\sharp + \frac{1}{2t} (\|x^{(k)} - x^\sharp\|_2^2 - \|x^{(k+1)} - x^\sharp\|_2^2)$$

6. Show that

$$\sum_{i=1}^k f(x^{(i)}) - v^\sharp \leq \frac{1}{2t} \|x^{(0)} - x^\sharp\|_2^2$$

7. Conclude that

$$f(x^{(k)}) - v^\# \leq \frac{1}{2kt} \|x^{(0)} - x^\#\|^2.$$

Exercise 4 (Kelley's convergence). We are going to prove that, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, and X a non-empty polytope (bounded polyhedron) then Kelley's cutting plane algorithm is converging.

Consider $x_1 \in X$. We consider a sequence of points $(x^{(k)})_{k \in \mathbb{N}}$ such that $x^{(k+1)}$ is an optimal solution to

$$(\mathcal{P}^{(k)}) \quad \underline{v}^{(k+1)} = \underset{x \in X}{\text{Min}} \quad z$$

$$\text{s.t.} \quad f(x^{(\kappa)}) + \langle g^{(\kappa)}, x - x^{(\kappa)} \rangle \leq z \quad \forall \kappa \in [k]$$

where $g^{(k)} \in \partial f(x^{(k)})$.

Denote $v = \min_{x \in X} f(x)$.

1. Show that v exists and is finite, and that there exists a sequence $x^{(k)}$.
2. Show that there exists L such that, for all k_1 and k_2 , we have $\|f(x^{(k_1)}) - f(x^{(k_2)})\| \leq L\|x^{(k_1)} - x^{(k_2)}\|$, and $\|g^{(k)}\| \leq L$.
3. Let $K_\varepsilon = \{k \in \mathbb{N} \mid f(x^{(k)}) > v + \varepsilon\}$ be the set of index such that $x^{(k)}$ is not an ε -optimal solution. Show that $f(x_k) \rightarrow v$ if and only if K_ε is finite for all $\varepsilon > 0$.
4. Consider $k_1, k_2 \in K_\varepsilon$, such that $k_2 > k_1$. Show that

$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v$$

5. Show that $\varepsilon + f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle < f(x^{(k_2)})$

6. Show that $\varepsilon < 2L\|x^{(k_2)} - x^{(k_1)}\|$.

7. Prove that $f(x^{(k)}) \rightarrow v$.

8. (Optional - hard) Find a complexity bound for the method (that is a number of iteration N_ε after which you are sure to have obtained a ε -optimal solution).